Answer Redistribution under Middle-Layer Replacement: An Answer-Identity Audit of Layer Composition

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Abstract

We show that, in this setting, direct middle-layer replacement can leave aggregate task accuracy nearly unchanged in point estimate while substantially redistributing which answers a language model gives on individual examples. We study this answer-redistribution phenomenon through direct middle-layer replacement in a multilingual reasoning setting, using Qwen2.5-1.5B-Instruct as the recipient model and Qwen2.5-1.5B Base as the donor. In the canonical Chinese MGSM setting, replacing layers 8–19 changes aggregate accuracy only modestly, from 0.500 to 0.468, yet changes the parsed answer on 151 of 250 examples (0.604; bootstrap 95% CI $[0.540, 0.664]$). A transition decomposition shows that the small accuracy delta arises from partially offsetting broken and repaired examples: 39 baseline-correct examples become wrong, while 31 baseline-wrong examples become correct. A refined decomposition shows that redistribution extends beyond correctness flips: 81 of 94 stable-wrong examples change from one incorrect numerical answer to another (0.862; bootstrap 95% CI $[0.787, 0.929]$). Composition-path controls reduce the plausibility of wrapper artifacts. Beyond these controls, behavioral diagnostics narrow simpler explanations: raw-prompt candidate-margin analyses do not support a simple final-answer preference flip, while numerical-trace divergence and rationale-conditioned scoring are consistent with a trajectory-level account. We treat these diagnostics as evidence about generated behavior, not as causal mediation, circuit identification, or reasoning-layer localization. Together, these results motivate tracking answer identity alongside aggregate accuracy when evaluating behavioral preservation under layer composition.

1 Introduction

Layer-composition interventions affect not only how often a model is correct, but also which answer the model produces on individual examples. We study this phenomenon as *answer redistribution*: large-scale movement in parsed answer identities under a structural model intervention. This perspective is important because a layer-composed model may appear nearly unchanged under aggregate accuracy while producing different answers on many individual examples.

We study this answer-redistribution phenomenon through direct middle-layer replacement in multilingual mathematical reasoning. In this layer-composition setup, we use Qwen2.5-1.5B-Instruct as the recipient model and Qwen2.5-1.5B Base as the donor, replacing layers 8–19 in the recipient with the corresponding base-model layers and evaluating the resulting direct-swap condition on Chinese MGSM ($n = 250$). The intervention changes aggregate accuracy only modestly, from 0.500 to 0.468 (accuracy delta -0.032; bootstrap 95% CI $[-0.096, 0.032]$), yet changes the parsed answer on 151 of 250 examples (0.604; bootstrap 95% CI $[0.540, 0.664]$). This small accuracy delta is produced by partially offsetting transitions: 39 baseline-correct examples become wrong, while 31 baseline-wrong examples become correct.

The redistribution is not limited to correctness flips. Among 94 examples that remain wrong under both conditions, only 13 preserve the same incorrect numerical answer; the remaining 81 move to a different incorrect answer (0.862; bootstrap 95% CI $[0.787, 0.929]$). Thus, even when the accuracy signal is small, the intervention often changes which answer the model produces. This stable-wrong decomposition is central to the audit: it shows that correctness alone can miss answer-identity movement inside the wrong subset.

We further evaluate several simpler explanations and behavioral diagnostics. Composition-path controls reduce the plausibility that the effect is caused by the wrapper itself, and repetition-filtered analyses show that the redistribution is not concentrated only in degenerate outputs. Raw-prompt candidate-margin analyses do not support a simple final-answer preference flip. Numerical-trace divergence and rationale-conditioned scoring are more compatible with a trajectory-level account of answer redistribution than with this alternative, but we treat these results as behavioral diagnostics rather than as evidence for causal mediation, reasoning-layer localization, or circuit identification.

Overall, we document answer redistribution under this direct middle-layer replacement setting, quantify it through parsed-answer change rate, transition decomposition, and stable-wrong refinement, and use controls and behavioral diagnostics to narrow simpler explanations without claiming causal-circuit identification.

2 Related Work

Layer composition, layer swapping, and model stitching. Layer-level composition has been studied as a way to analyze, combine, or transfer model capabilities. Model stitching connects the bottom layers of one trained model to the top layers of another, often with a trainable stitching layer, and uses the resulting performance to compare internal representations or functional compatibility between networks [Bansal et al., 2021]. A related line of model-composition work studies whether independently fine-tuned models or parameter updates can be combined through weight averaging, task vectors, or related merging operations [Wortsman et al., 2022, Ilharco et al., 2023]. These methods show that model components or parameter updates can sometimes be recombined while preserving useful behavior.

Layer swapping provides a more direct form of structural composition. Bandarkar et al. [2025] swap transformer layers between math- and language-specialized fine-tunes to improve zero-shot cross-lingual mathematical transfer. Related layer-wise swapping methods have also been explored for multilingual safety alignment [Shin and Hwang, 2026]. Our work uses direct middle-layer replacement differently. We replace middle layers between an instruction-tuned model and its base-model sibling, and we audit answer-level redistribution rather than transfer, safety, or aggregate performance improvement.

Aggregate metrics, prediction churn, and hidden instance-level change. Predictive multiplicity and related work establish that similar aggregate performance can coexist with substantial instance-level differences. In classification, multiple competing models can achieve nearly equivalent accuracy while disagreeing on individual predictions [Marx et al., 2020]. Work on underspecification similarly argues that many distinct models may obtain similar held-out performance while relying on different internal solutions or behaving differently under deployment conditions [D'Amour et al., 2022].

Prediction churn provides a particularly close motivation for our setting. Model updates can preserve or improve aggregate accuracy while changing individual predictions, creating instability that is invisible from accuracy alone [Milani Fard et al., 2016]. Our study transfers this concern to layer-composition interventions in language models. We do not compare independently trained classifiers or successive production models. Instead, we compare a language model before and after a controlled middle-layer replacement. The relevant question is therefore not only whether two systems have similar accuracy, but whether the intervention preserves the identity of the parsed answers it produces.

Intervention-based interpretability and activation patching. Intervention-based interpretability methods also motivate our analysis, but our claims are more limited. Causal mediation, causal tracing, activation patching, and related interchange-intervention frameworks intervene on hidden states or model components to estimate their effect on outputs and to test hypotheses about internal computation [Vig et al., 2020, Meng et al., 2022, Geiger et al., 2021, Zhang and Nanda, 2024]. These methods are often used to support localization or mechanistic claims, but their conclusions can depend strongly on metrics, corruption choices, and intervention design [Heimersheim and Nanda, 2024].

Our direct middle-layer replacement is also an intervention, but we do not treat it as causal-circuit identification or reasoning-layer localization. It is a coarse structural perturbation used to expose answer redistribution under layer composition. The patching analyses in our experiments serve a narrower role: they test whether downstream states can change outputs toward the baseline answer in a source-conditioned way, not whether a complete reasoning mechanism has been isolated. This distinction is especially important for mathematical reasoning, where activation-level interventions have been used to study arithmetic computations, but such analyses require stronger causal designs than the diagnostic claims made here [Stolfo et al., 2023].

Chain-of-thought faithfulness and trajectory diagnostics. Our trajectory-level diagnostics are related to work on chain-of-thought reasoning and rationale faithfulness. Chain-of-thought prompting can improve reasoning performance on multi-step tasks [Wei et al., 2022], but generated rationales are not guaranteed to faithfully represent the model's internal reasoning process [Turpin et al., 2023, Lanham et al., 2023]. This caveat is central to how we interpret our rationale-conditioned scoring and numerical-trace analyses.

We therefore treat these analyses as behavioral diagnostics rather than as faithful mechanistic explanations. In our setting, generated rationales and numerical traces help characterize the form of answer redistribution and narrow simpler explanations, but they do not prove causal mediation from a generated rationale to the final answer. This positioning lets us use trajectory-level evidence without assuming that the observed rationale is the model's true internal reasoning process.

Position of this work. Prior work suggests three relevant lessons: layer composition can preserve or transfer useful behavior, aggregate metrics can hide instance-level disagreement, and intervention-based interpretability requires careful causal designs. We connect these concerns by studying answer redistribution under direct middle-layer replacement. Thus, we position this paper as an answer-redistribution study of layer composition: direct middle-layer replacement can leave aggregate accuracy nearly unchanged while substantially reassigning individual parsed answers.

3 Approach

3.1 Direct Middle-Layer Replacement

We study direct middle-layer replacement between a base model and its instruction-tuned sibling. Let M^{inst} denote the instruction-tuned recipient model and M^{base} denote the base-model donor. Both models have the same transformer architecture and the same number of decoder layers. For a boundary pair (b, t), we construct a direct-swap model $M^{\mathrm{swap}}_{b:t}$ by taking the bottom layers $0, \dots, b-1$ from M^{inst}, the middle layers $b, \dots, t-1$ from M^{base}, and the top layers $t, \dots, L-1$ from M^{inst}.

Our canonical intervention uses $b = 8$ and $t = 20$ in Qwen2.5-1.5B, which has $L = 28$ decoder layers. Thus, layers 8–19 are replaced by the corresponding base-model layers, while layers 0–7 and 20–27 remain from the instruction-tuned recipient. Both conditions use the same prompt format, decoding configuration, and evaluation pipeline; full generation details are provided in Section 4. We use this intervention as a fixed structural perturbation, not as a training method or model-improvement algorithm. The goal is to test whether a layer-composed model that preserves aggregate accuracy also preserves answer identity on individual examples.

3.2 Answer-Level Transition Analysis

For each example i, we generate an output from the baseline recipient model and from the direct-swap model. We extract a numerical answer from each output using the same answer parser across conditions, and we refer to this normalized value as the parsed answer. Correctness is evaluated by comparing the parsed answer with the gold MGSM answer.

Crucially, we use the same normalized-answer policy for both correctness evaluation and answer-identity comparison. Parsed numerical answers are normalized before comparison, so decimal-equivalent strings such as "75.00" and "75" are treated as equivalent. Parse failures are retained as incorrect. Let a_i^{base} and a_i^{swap} denote the normalized parsed answers from the baseline and direct-swap conditions. We define the answer-change indicator as

$$\Delta_{\text{ans}}(i) = \mathbb{1}\left[a_i^{\text{base}} \neq a_i^{\text{swap}}\right],$$

and report the parsed-answer change rate as the mean of $\Delta_{\text{ans}}(i)$ over the evaluation set. This metric captures whether the intervention changes the answer identity, regardless of whether either answer is correct. Confidence intervals for paired comparisons are computed by paired bootstrap over sample IDs; implementation details appear in Section 4.

Aggregate accuracy alone cannot reveal how individual examples move between correct and incorrect answers. We therefore decompose examples into four transition groups based on correctness before and after the intervention:

$$\begin{aligned}
S_{\text{stable_correct}} &= \{i : c_i^{\text{base}} = 1,\ c_i^{\text{swap}} = 1\},\\
S_{\text{broken}} &= \{i : c_i^{\text{base}} = 1,\ c_i^{\text{swap}} = 0\},\\
S_{\text{repaired}} &= \{i : c_i^{\text{base}} = 0,\ c_i^{\text{swap}} = 1\},\\
S_{\text{stable_wrong}} &= \{i : c_i^{\text{base}} = 0,\ c_i^{\text{swap}} = 0\},
\end{aligned}$$

where c_i^{base} and c_i^{swap} indicate correctness under the baseline and direct-swap conditions.

This decomposition separates accuracy loss from accuracy gain. The accuracy delta is determined by the balance between S_{broken} and S_{repaired}, while answer redistribution can remain hidden inside correctness-preserving groups, especially $S_{\text{stable\_wrong}}$. Because mathematical answers are parsed as numerical identities, examples may remain incorrect while changing from one wrong answer to another.

We therefore refine the stable-wrong group into same-answer and different-answer subsets:

$$\begin{aligned}
S_{\text{stable_wrong_same}} &= \{i \in S_{\text{stable_wrong}} : a_i^{\text{base}} = a_i^{\text{swap}}\},\\
S_{\text{stable_wrong_diff}} &= \{i \in S_{\text{stable_wrong}} : a_i^{\text{base}} \neq a_i^{\text{swap}}\}.
\end{aligned}$$

This refinement is central to our audit because correctness alone treats both subsets identically, even though the second subset reflects answer-identity redistribution.

3.3 Controls and Diagnostic Analyses

We use several controls and diagnostics to distinguish answer redistribution from simpler artifacts. First, we include an identity-composition control in which the layer-composition wrapper is executed without changing the effective model. This control tests whether answer changes are caused by the wrapper or generation harness rather than by replacing middle layers. Second, we use repetition and output-mode analyses to check whether the redistribution is concentrated only in degenerate generations or parse failures.

We also use behavioral diagnostics to evaluate simpler explanations of the answer changes. A raw-prompt candidate-margin analysis scores the baseline and direct-swap candidate answers under each condition from the raw prompt, without conditioning on the generated rationale. If answer redistribution were primarily a simple final-answer preference flip, we would expect the direct-swap condition to assign systematically higher relative likelihood to its own final answer from the raw prompt alone.

To test whether answer changes are more closely associated with generated trajectories, we compute numerical-trace divergence between baseline and direct-swap outputs. This diagnostic compares extracted numerical-token sequences from the two generations using normalized edit distance. We also use rationale-conditioned scoring, which evaluates candidate final-answer likelihoods when conditioned on generated rationales from different conditions. These diagnostics are used to characterize generated behavior and to contrast static final-answer preference changes with trajectory-level differences in generated reasoning traces. We discuss the evidential scope of this trajectory-level interpretation in Section 5.2.

4 Experiments

4.1 Data

Our main evaluation uses Chinese MGSM [Shi et al., 2023], a multilingual extension of GSM8K-style grade-school mathematical reasoning problems [Cobbe et al., 2021]. The main setting contains $n = 250$ Chinese MGSM examples. Each example requires the model to generate a solution and produce a final numerical answer. We also evaluate Korean and Arabic MGSM with $n = 250$ examples each.

4.2 Evaluation method

We evaluate each generated solution by extracting a numerical answer and comparing it with the gold MGSM answer. The same parser and normalized-answer policy are used for correctness evaluation and answer-identity comparison. Decimal-equivalent parsed answers, such as "75.00" and "75", are treated as equivalent. Parse failures are counted as incorrect.

We report three primary quantities. First, we report aggregate accuracy for the baseline and direct-swap conditions. Second, we report the parsed-answer change rate, defined in Section 3 as the mean of $\Delta_{\mathrm{ans}}(i)$ over the evaluation set. Third, we report transition-group counts, as defined in Section 3, including stable-correct, broken, repaired, stable-wrong, and the stable-wrong same-answer/different-answer refinement.

Unless otherwise noted, 95% confidence intervals are computed by paired nonparametric bootstrap over sample IDs with 10,000 resamples using bootstrap seed 20260517. Each bootstrap sample resamples example IDs and keeps the corresponding baseline and direct-swap outputs together. These intervals quantify sample-level uncertainty under the fixed decoding protocol. They do not quantify variability over alternative prompt formats, model families, layer ranges, or decoding strategies.

4.3 Experimental details

We use Qwen2.5-1.5B as the model family for all experiments [Qwen Team, 2025]. The baseline recipient is Qwen2.5-1.5B-Instruct, and the donor is Qwen2.5-1.5B Base. These models share the same decoder-only transformer architecture and number of layers, allowing direct layer replacement without trainable adapters or projection modules.

The direct-swap condition replaces layers 8–19 of Qwen2.5-1.5B-Instruct with the corresponding layers from Qwen2.5-1.5B Base. The unmodified Qwen2.5-1.5B-Instruct model is the baseline condition. Unless otherwise stated, all comparisons are made between these two conditions under the same prompt format, decoding configuration, and evaluation pipeline.

All reported Chinese, Korean, and Arabic Qwen evaluations use a raw prompt format rather than a chat template. The baseline and direct-swap conditions are decoded greedily with `do_sample=false`, temperature 0.0, and a maximum length of 256 new tokens. Because these generations are deterministic, the paired answer-change estimates are not driven by differences in sampling streams across conditions.

Outputs are compared at the sample level: for each example, the baseline and direct-swap outputs form a paired comparison. The identity-composition control uses the same harness while leaving the effective model unchanged.

We also perform source-specific patch-control audits on the broken set S_{broken}, that is, examples that are correct under the baseline condition but incorrect under direct-swap. These audits compare same-sample baseline-state patches with shuffled baseline-state and norm-matched random controls at downstream layers 20 and 22, shortly after the replaced layer range. For the patch-control comparisons, paired bootstrap confidence intervals are computed over the same S_{broken} sample IDs using 10,000 resamples and bootstrap seed 20260517.

We additionally analyze whether the main redistribution effect is concentrated in degenerate or repetitive direct-swap outputs. We report a repetition-filtered subset analysis in Appendix C as a robustness check.

Finally, we use behavioral diagnostics—raw-prompt candidate-margin analyses, numerical-trace divergence using normalized edit distance, and rationale-conditioned scoring—to compare static final-answer preference explanations with trajectory-level explanations. These diagnostics are interpreted as behavioral evidence about generated outputs, not as causal mediation or circuit-identification evidence.

4.4 Results

The direct middle-layer replacement produces only a modest change in aggregate accuracy, but a much larger change in answer identity. On Chinese MGSM, the baseline Qwen2.5-1.5B-Instruct model obtains 0.500 accuracy, while the direct-swap condition obtains 0.468 accuracy. The resulting accuracy delta is -0.032 with a bootstrap 95% confidence interval of $[-0.096, 0.032]$.

This small aggregate change masks substantial answer redistribution. The parsed answer changes on 151 of 250 examples, corresponding to a parsed-answer change rate of 0.604 with a bootstrap 95% confidence interval of $[0.540, 0.664]$. Thus, even though the aggregate accuracy difference is small, the intervention changes which answer the model produces on most examples.

The transition decomposition explains why the accuracy delta is small. Among the 250 examples, 86 remain correct under both conditions, 39 are broken by the intervention, 31 are repaired by the intervention, and 94 remain wrong under both conditions. The 39 broken examples reduce accuracy, while the 31 repaired examples partially offset this loss. As a result, aggregate accuracy understates the amount of answer movement induced by the intervention.

The stable-wrong group shows that the redistribution is not limited to correctness flips. Among the 94 examples that remain wrong under both conditions, only 13 preserve the same incorrect numerical answer. The remaining 81 move from one incorrect answer to another, corresponding to a stable-wrong different-answer rate of 0.862 with a bootstrap 95% confidence interval of $[0.787, 0.929]$. This result is central to the audit: correctness alone treats all stable-wrong examples identically, but answer identity reveals substantial movement inside the wrong subset. Full transition decompositions for both languages are reported in Appendix A.

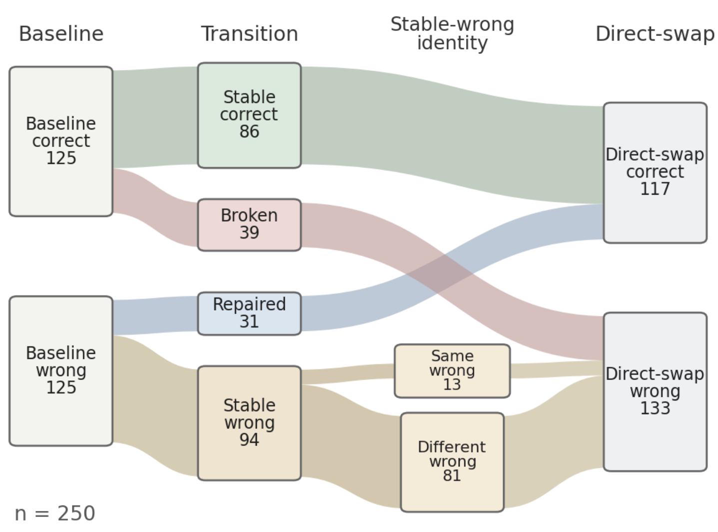

Figure 1: Transition decomposition for Chinese MGSM under direct middle-layer replacement (n = 250). Baseline and direct-swap correctness counts are connected by transition groups (Stable correct, Broken, Repaired, Stable wrong), and the Stable wrong group is further refined into Same-wrong and Different-wrong subsets. Aggregate accuracy changes only modestly (125 $\rightarrow$ 117 correct), while most stable-wrong examples move to a different incorrect numerical answer.

The additional-language sanity checks show the same qualitative pattern. On Korean MGSM, baseline accuracy is 0.312 and direct-swap accuracy is 0.320, while the parsed answer changes on 178 of 250 examples, corresponding to a parsed-answer change rate of 0.712. On Arabic MGSM, baseline accuracy is 0.224 and direct-swap accuracy is 0.272, while the parsed answer changes on 183 of 250 examples, corresponding to a parsed-answer change rate of 0.732. Across all three languages, aggregate accuracy changes are small relative to the rate of parsed-answer changes. These results suggest that the redistribution pattern is not unique to Chinese, although they do not establish broad generalization across model families or task types.

5 Analysis

5.1 Source-Specific Patch Controls

The transition results show that direct middle-layer replacement substantially redistributes parsed answers. We next ask whether downstream states can change direct-swap outputs toward the baseline answer in a source-conditioned way. Using the patch-control protocol described in Section 4, we analyze the $n = 39$ broken examples in S_{broken}.

The same-sample patches redirect outputs toward the baseline answer much more often than the control patches. At layer 20, same-sample baseline-state patches match the baseline answer on 19 of 39 examples, while shuffled baseline-state patches and norm-matched random controls each match the baseline answer on 0 of 39 examples. This yields a same-sample minus shuffled difference of 0.487 with a bootstrap 95% confidence interval of $[0.333, 0.641]$, and a same-sample minus random difference of 0.487 with the same confidence interval.

At layer 22, same-sample baseline-state patches similarly match the baseline answer on 19 of 39 examples. Shuffled baseline-state patches match the baseline answer on 1 of 39 examples, and norm-matched random controls match the baseline answer on 2 of 39 examples. The same-sample minus shuffled difference is 0.462 with a bootstrap 95% confidence interval of $[0.308, 0.615]$, and the same-sample minus random difference is 0.436 with a bootstrap 95% confidence interval of $[0.282, 0.590]$.

Layer	Condition	Match baseline	Match direct-swap	Other	Parse fail
–	No patch	0/39	39/39	0/39	0/39
20	Same-sample baseline patch	19/39	8/39	12/39	0/39
20	Shuffled baseline patch	0/39	1/39	38/39	0/39
20	Random norm-matched	0/39	1/39	33/39	5/39
22	Same-sample baseline patch	19/39	5/39	15/39	0/39
22	Shuffled baseline patch	1/39	4/39	34/39	0/39
22	Random norm-matched	2/39	0/39	36/39	1/39

Table 1: Patch-control results on the broken set S_{broken} ($n = 39$). Same-sample baseline-state patches at layers 20 and 22 change direct-swap outputs toward the baseline answer more often than shuffled or norm-matched random controls.

These patch-control results reduce the plausibility that the effect is caused by generic perturbation magnitude alone, supporting a source-conditioned output change pattern. However, they do not identify a complete reasoning mechanism or localize the source of the answer change.

5.2 Behavioral Diagnostics

We use behavioral diagnostics to compare a simple final-answer preference explanation with a non-causal trajectory-level description of answer redistribution.

The raw-prompt candidate-margin analysis does not support a simple final-answer preference flip. On the 151 answer-changed examples, the mean margin delta is 0.0065 with a bootstrap 95% confidence interval of $[-0.0428, 0.0551]$.

The second diagnostic compares generated numerical trajectories. Answer-changed examples show substantially larger numerical-trace divergence than answer-unchanged examples. The mean normalized edit distance between extracted numerical-token sequences is 0.700 for answer-changed examples and 0.448 for answer-unchanged examples, yielding a difference of 0.252 with a bootstrap 95% confidence interval of $[0.205, 0.300]$. This suggests that answer changes are accompanied by changes in the generated numerical trajectory, rather than appearing only as isolated final-answer substitutions.

The third diagnostic uses rationale-conditioned scoring. We score candidate final answers under combinations of model condition and generated rationale source. As shown in Table 2, final-answer margins track the generated rationale source more strongly than the model condition: both models favor the baseline answer when conditioned on the baseline rationale and favor the direct-swap answer when conditioned on the direct-swap rationale.

Model condition	Rationale source	Mean margin	95% CI
Baseline	Baseline rationale	$+3.20$	$[2.894, 3.487]$
Baseline	Direct-swap rationale	-2.11	$[-2.438, -1.774]$
Direct-swap	Baseline rationale	$+2.69$	$[2.446, 2.935]$
Direct-swap	Direct-swap rationale	-1.90	$[-2.187, -1.616]$

Table 2: Rationale-conditioned scoring on answer-changed examples. Final-answer margins track the generated rationale source more strongly than the model condition. We interpret this as behavioral evidence, not as causal mediation.

Together, these diagnostics narrow a simple final-answer preference explanation. Raw-prompt candidate-margin analysis does not show a systematic direct-swap preference from the prompt alone, while numerical-trace divergence and rationale-conditioned scoring indicate that answer changes are associated with changes in the generated trajectory. These results are more consistent with a trajectory-level behavioral description than with a simple raw-prompt final-answer preference flip, subject to the evidential limits discussed below.

Evidential scope of the trajectory-level interpretation. We use the term *trajectory-level account* in a deliberately non-causal sense. It refers to an associational pattern in which answer changes are better characterized by changes in the generated reasoning trajectory than by a static final-answer preference measured from the raw prompt alone. This does not establish that the generated trajectory causally mediates the final answer: a shared factor, such as an internal state that affects both the trajectory and the final answer, could explain the observed pattern.

Our trajectory diagnostics are also derived from generated text and may therefore share output-level confounds, including length or formatting differences. Consistent with this caution, our hidden-state probe audit in Appendix D finds that a length-only control achieves higher discrimination than the hidden-state divergence features considered in that audit. Stronger mediation claims would require controlled trajectory interventions, which we leave to future work.

6 Conclusion

This report examined answer redistribution under direct middle-layer replacement in layer-composition interventions. The main Chinese MGSM result shows that a small aggregate accuracy change can coexist with extensive reassignment of individual parsed answers. The transition analysis explains this pattern: broken and repaired examples partially offset in accuracy, while most stable-wrong examples move to different incorrect numerical answers.

Auxiliary analyses narrow several simpler explanations. Identity-composition and output-mode controls reduce the plausibility of wrapper artifacts or degenerate-output effects. Source-specific patch controls show that same-sample baseline states change direct-swap outputs toward the baseline answer more often than shuffled or random controls. Behavioral diagnostics further suggest that answer changes are better characterized by generated-trajectory differences than by a simple raw-prompt final-answer preference flip. However, we do not claim to identify a causal circuit, localize mathematical reasoning to specific layers, or prove that generated rationales causally mediate final answers.

The evidence is limited to one model pair, one task family, a fixed greedy decoding protocol, and a boundary choice informed by exploratory sweeps; Korean and Arabic sanity checks suggest that the qualitative pattern is not unique to Chinese but do not establish broader generalization. The narrower conclusion is that direct middle-layer replacement can induce large-scale answer redistribution without a commensurate aggregate accuracy drop. When behavioral preservation matters, layer-composition studies should therefore report not only aggregate accuracy, but also whether individual answer identities are preserved. Even examples that remain incorrect can reveal redistribution invisible to correctness alone.

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A Additional Language Results

We report Korean and Arabic MGSM sanity checks to assess whether the qualitative answer-redistribution pattern is unique to Chinese. These runs use the same model family, direct-swap construction, answer parsing policy, and paired comparison setup as the main Chinese MGSM experiment. They are intended as sanity checks, not as evidence for broad cross-lingual generalization across model families, model scales, or task types.

Language	Baseline acc.	Direct-swap acc.	Answer changed	Stable correct	Broken	Repaired	Stable wrong
Korean	0.312	0.320	178/250 (0.712)	54	24	26	146
Arabic	0.224	0.272	183/250 (0.732)	38	18	30	164

Table 3: Additional-language sanity checks on Korean and Arabic MGSM. Both languages show large parsed-answer change rates despite small aggregate accuracy changes.

The stable-wrong refinement shows that the same qualitative pattern also appears inside the wrong subset. For Korean, 128 of 146 stable-wrong examples move to a different incorrect numerical answer, corresponding to a stable-wrong different-answer rate of 0.877. For Arabic, 135 of 164 stable-wrong examples move to a different incorrect numerical answer, corresponding to a stable-wrong different-answer rate of 0.823.

Language	Stable-wrong total	Stable-wrong same	Stable-wrong different
Korean	146	18	128/146 (0.877)
Arabic	164	29	135/164 (0.823)

Table 4: Stable-wrong refinement for the additional-language sanity checks. Even when examples remain incorrect under both conditions, most move to a different incorrect numerical answer.

These results support a narrow robustness conclusion. The Chinese result is not an isolated artifact of one language subset: Korean and Arabic also show small aggregate accuracy differences alongside large parsed-answer changes. However, because all three settings use the same Qwen2.5-1.5B Base/Instruct pair and MGSM-style mathematical reasoning tasks, these checks do not establish generality across model families or task domains.

B Patch-Control Sanity Details

The main patch-control analysis in Section 5 compares same-sample baseline-state patches against shuffled baseline-state and norm-matched random controls on the broken set S_{broken}. This appendix reports an additional harness sanity check using direct-swap self-patching. In this condition, direct-swap states are patched back into the direct-swap run at the same downstream layers. The purpose is to check whether the patching harness approximately preserves direct-swap behavior when the patched state comes from the same direct-swap trajectory.

Layer	Condition	Match baseline	Match direct-swap	Other	Parse fail
–	No patch	0/39	39/39	0/39	0/39
20	Direct-swap self-patch	0/39	38/39	1/39	0/39
22	Direct-swap self-patch	0/39	38/39	1/39	0/39

Table 5: Patch-control harness sanity check on S_{broken} ($n = 39$). Direct-swap self-patching reproduces the direct-swap answer in nearly all cases, supporting the interpretation that the patching harness itself is not responsible for the baseline-directed redirection observed under same-sample baseline-state patches.

This sanity check is interpreted narrowly. It does not provide evidence for a reasoning circuit, a causal mechanism, or layer-localized reasoning. It only verifies that patching direct-swap states into the direct-swap computation approximately preserves direct-swap output behavior. The main patch-control evidence therefore remains the contrast between same-sample baseline-state patches and the shuffled or norm-matched random controls reported in Table 1.

C Output-Mode and Repetition Robustness

We check whether the main answer-redistribution effect is concentrated in repetitive, degenerate, or otherwise flagged direct-swap outputs. This robustness check removes 24 examples whose direct-swap outputs are flagged by the output-mode and repetition checks, and recomputes the main answer-identity metrics on the remaining 226 examples.

Subset	n	Accuracy delta	Answer changed	Stable-wrong different
All examples	250	-0.032	151/250 (0.604)	81/94 (0.862)
Excluding flagged outputs	226	-0.031	135/226 (0.597)	70/83 (0.843)

Table 6: Robustness to excluding flagged direct-swap outputs. The parsed-answer change rate and stable-wrong different-answer rate remain high after excluding outputs flagged by output-mode and repetition checks.

The filtered subset preserves the main qualitative pattern. After excluding flagged outputs, the accuracy delta remains small at approximately -0.031, while the parsed-answer change rate remains high at 135/226 = 0.597. Within the stable-wrong subset, 70 of 83 examples still move to a different incorrect numerical answer, corresponding to a stable-wrong different-answer rate of 0.843.

This robustness check does not replace the full evaluation set, but it reduces the plausibility that the main redistribution effect is driven only by repetitive generations or degenerate output modes. We therefore retain the full set of 250 examples as the main analysis and report the filtered subset as an auxiliary robustness check.

D Hidden-State Divergence Audit

We include a hidden-state divergence audit as appendix-level association evidence. The purpose is to test whether examples whose parsed answers change are weakly separable from answer-stable examples using hidden-state divergence features. This audit is not used as causal evidence, and we do not interpret it as reasoning-layer localization.

Under the final normalized-answer evaluation, there are 151 answer-changed examples and 99 answer-unchanged examples. We compute layer-wise L2 hidden-state divergence features between the baseline and direct-swap conditions and evaluate how well these features distinguish answer-changed from answer-unchanged examples. The best single-layer feature is the layer-10 L2 divergence, with ROC-AUC 0.679. An all-layer hidden-state probe obtains ROC-AUC 0.637. However, a length-only control obtains ROC-AUC 0.753, exceeding the hidden-state divergence features considered in this audit.

Diagnostic	ROC-AUC
Best single-layer L2 divergence, layer 10	0.679
All-layer hidden-state probe	0.637
Length-only control	0.753

Table 7: Hidden-state divergence audit for answer-changed versus answer-unchanged examples. Length-only controls outperform the hidden-state divergence features considered here, so we treat this result as weak association evidence rather than localization or causal evidence.

The length-only control is an important caveat. It suggests that hidden-state divergence signals may be entangled with output-level factors such as generation length. For this reason, we do not use this audit to claim that layer 10 is a reasoning layer or that hidden-state divergence causally explains answer redistribution. The audit only supports the weaker observation that answer-changed examples differ from answer-stable examples in ways that are partly visible in internal-state-derived features, while also being strongly confounded by length-related signals.